



# Role of nanoparticles in fish disease management: A review

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## ARTICLE INFO

Handling editor: Ching Hou

### Keywords:

Nanoparticles  
Fish disease  
Pathogen  
Bacterium  
Parasite  
Control

## ABSTRACT

Aquaculture is undoubtedly a flourishing industry that ensures food security for the increasing population. However, this industry is under severe threat from bacterial, viral, parasitic, and fungal infections besides environmental contamination. Hence, in the recent days, new technologies are surfacing to confront these challenges efficiently. Nanotechnology is one of the novel and pioneering skills with a wide continuum of applications ranging from aquatic pond sterilization, water treatment, nutrient delivery, diagnosis, drug delivery, aquatic disease control, etc. The present review is mainly focused on the application of nanomaterials in fish disease management to promote sustainable aquaculture. Specifically, the effects of gold, silver, copper oxide, zinc oxide, selenium, chitosan, platinum, etc., nanoparticles on pathogen identification and treatment of fish diseases are discussed in detail. In addition, the role of these nanoparticles in fish disease control is highlighted in this review. In contrast, this review also emphasizes the toxicity associated with nanoparticle-based fish disease management which will be helpful for fish culturists.

## 1. Introduction

It is estimated that the world's population will reach 9.7 billion by 2050 and to meet the global demand, the food production has to increase 70–100% from the current generation level (Mohanty, 2015; Gudivada and Vankara, 2020). Fishes are considered to be an indispensable source of proteins and about 17% of the world's population is dependent on fish due to their key role in food safety, nutrition, and income (Shah and Mraz, 2020). They are a high-quality food source that is rich in necessary nutrients like essential amino acids, fatty acids [(Polyunsaturated fatty acids (PUFAs) and omega 3-fatty acids such as eicosapentaenoic acid (EPA), and docosahexaenoic acid (DHA)], vitamins (A, C, D, E, K, and B complex), and minerals (iron, calcium, phosphorous, selenium and fluorine) (Daniel, 2016; Nasr-Eldahan et al., 2021). Due to affordable and widely available sources of animal protein, fish is widely considered as poor man's food around the world. In addition, fish will be a prominent future contributor to the global food supply, and has been envisioned that another supplementary 27 million tonnes of food fish have to be produced globally by 2030 to maintain its consumption at the unseen status (Modi et al., 2023). The preliminary estimation showed that there is a steady increase in the global annual per capita consumption of fish from 9.9 kg (1960) to more than 20 kg (2015) (Sherif et al., 2023). In order to meet the demand, aquaculture production should increase double in size which can be met from farmed fish supplies.

Fishes are acknowledged to lessen the danger of coronary heart disease (CHD) deaths apart from several other diseases. Moreover, fish is helpful for brain health, curing asthma, lessen the depression, prevent autoimmune diseases etc. (Vankara et al., 2022a; Mandal

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et al., 2024). Although a major portion of the population benefits from fishery, it has a threat from bacterial, viral, parasitic, and fungal infections which affects fish productivity. This leads to food insecurity, loss of income, and ecological balance (Abinaya et al., 2023). The aquatic environment will often be contaminated by these pathogens besides an increased output of aquaculture waste is reported. The damage caused by infections with various pathogens is well recorded from both marine and freshwater fishes (Riaz et al., 2023). Hence, there is an urgent need to solve such kind of outbreaks in the aquaculture industry and take precautionary measures to avoid further damage (Vankara et al., 2022b; Khursheed et al., 2023).

Nanotechnology opens enormous opportunities in different fields including the aquaculture sector. Nanoparticles (NPs) have unique physical and chemical properties such as high surface zone, high reactivity, tunable pore size, and particle morphology which are useful in many industrial applications (Varaprasad et al., 2022). Specifically, nanotechnology has evolved as a new avenue to solve the issues raised in the sustainability of aquaculture sectors (Abinaya et al., 2023). The fascination with nanotechnology is a result of the intriguing and extraordinary qualities demonstrated by nanoparticles, including their physical, chemical and biological activities, and their expanded applicability when compared to their bulk components (Pati and Mondal, 2019; Joudeh and Linke, 2022). Several research works proved the role of different nanoparticles in fish-pathogen interactions (Shaalán et al., 2016). Rapid development of metal and metal-oxide nanoparticles and their usage in several fields including electronics, energy, biomedicine etc., is a frequent practice in recent times besides understanding their nature (Liu et al., 2016; Khan et al., 2024). In general, metal-oxide nanoparticles are more popular due to their high reactivity and effectivity than metal NPs. Specifically, this review sheds light on the effects of nanoparticles on fish disease management. The mode of action of various nanoparticles makes them desirable as an antimicrobial tool in fish disease management.

Gold nanoparticles (Au NPs) are typically prepared in a wide variety of shapes such as spherical, tube, rods, etc., and possess size and shape-tunable properties (Saleh et al., 2016; Pierre et al., 2024). Gold nanoparticles have typical chemical, thermal, mechanical, electrical and optical properties. Gold NPs can be detected by magnetic resonance imaging and their surface electrons show plasmon resonance properties. Hence, these are used as contrast agents on multiple imaging platforms and used in biomedical research (Luo et al., 2021). Their unique chemical and physical properties make them promising candidates in drug target identification (Tu et al., 2020; Chen and Feng, 2022). Gold NPs are inert and non-toxic, and thus have gained attention in cancer research including drug delivery etc., and the combination of  $\text{Au}^{2+}$  and  $\text{Au}^{3+}$  proved to enhance the activity of antitumor drugs (Hassantabar et al., 2021; Huang et al., 2023). Silver nanoparticles (Ag NPs) are the major class of metal-NPs which possess properties like high conductivity, chemical stability, catalytic activity, scattering and nonlinear optical performance (Khan et al., 2015; Doszpoly et al., 2023). Silver NPs are extremely prevalent candidates in a broad range of materials and are used for food, medical, and other industrial purposes (Shaalán et al., 2017). In addition, Ag NPs are widely used to restrict the microbial growth and reduce water pollution (Ale et al., 2018; Parashar et al., 2022). Specifically, silver nanoparticles are used for drug delivery, topical and nano-medical applications (Ghetas et al., 2022). Khursheed et al. (2023) emphasized that silver nanoparticles are best alternatives to antimicrobial agents. The role of silver NPs as antibacterial, antifungal, antiviral, anti-inflammatory, anti-angiogenic, and anticancer agents were well established by several researchers (Mabrouk et al., 2021; Aliakbari et al., 2024).

Copper is an essential trace element found in food nutrients or supplements (Smriti et al., 2023). Similarly, copper nanoparticles (Cu NPs) are useful for food, biomedical and various other industrial applications (Woźniak-Budych et al., 2023). At low concentration, copper NPs are non-toxic to several organisms including aquatic species (Delavari et al., 2022). Moreover, copper nanoparticles and Cu NPs fused with other elements showed effective antimicrobial activities (Zhang et al., 2021). Copper oxide nanoparticles (CuO NPs) are also largely used in many industries such as food, paints, ceramics, electronics, pharmaceutical, biomedical, etc. (Swain et al., 2014; Letchumanan et al., 2021). Copper oxide nanoparticles are used as biocides and also as antimicrobial agents (Malhotra et al., 2020). Specifically, at low dosage, CuO nanoparticles are useful in antimicrobial activities (Balasubramanian et al., 2024). Zinc oxide nanoparticles (ZnO NPs) serve as a source for plastic, ceramic, glass, paint, and food industries (Shaalán et al., 2017). Zinc oxide nanoparticles also have the potential to diagnose and control various diseases (Jiang et al., 2018; Abinaya et al., 2023). In addition, zinc oxide nanoparticles stand as an example for biocompatible and biodegradable nanoplatforms that are promising in reducing the side effects (Awad et al., 2019). Although ZnO is helpful as a food supplement at low concentrations, it becomes toxic when over accumulated (Pati et al., 2021). Zinc oxide NPs are widely used in drug delivery systems due to their low toxicity and beneficial qualities (Rasheed et al., 2023). The role of ZnO NPs in antimicrobial, anticancer, anti-inflammatory and anti-diabetic activities including immunomodulatory effects was studied by various researchers (Lallo da Silva et al., 2019; Mandal et al., 2024).

Selenium is an essential trace element for microbial organisms and animals including fish and is available in both inorganic and organic forms. Selenium nanoparticles (Se NPs) are used in various fields due to their biocompatibility, bioavailability and low toxicity (Dawood et al., 2021). These nanoparticles fight against fungal, bacterial, and parasitic infections including diabetes and cancer (Dash et al., 2022). Selenium functions as an excellent antioxidant and promotes a healthy immune system. Also, Se NPs act as chemopreventive and anti-inflammatory agents (Yazhiniprabha et al., 2022). Different sources of selenium NPs boost final weight, relative gain rate, and antioxidant status (Zhu et al., 2023). Inclusion of selenium NPs avoid cell damage and play active role in ROS protection, growth, fertility and immunity of fish (Ni et al., 2023).

Chitosan [(1, 4)-2-amino-2-deoxy-D-glucan] is a linear polysaccharide which is an abundant natural bio-polymer like cellulose. When chitin undergoes deacetylation, it turns into chitosan, and the extent of deacetylation affects its properties and applications (Abdel-Razek, 2019). Chitosan nanoparticles (CS NPs) find various uses across agriculture, food protection, cosmetics, medicine, pharmaceuticals, drug delivery, drug formulations, and wastewater treatment (Bhoopathy et al., 2021). Chitosan NPs popularity in these fields is mainly due to its biocompatibility and biodegradability and low toxicity (El-Naggar et al., 2022; Zhang et al., 2023). Chitosan has antimicrobial properties which includes antibacterial, antifungal, antiviral and antiparasitic activities (Khan et al., 2024). Although in recent years, many studies reported aquatic fate with various nanoparticles, only a few studies have concentrated

on the effects of platinum nanoparticles (Pt NPs). The platinum NPs are more relevant due to their extensive usage in various industries (Lakshmi et al., 2023). Platinum NPs have antimicrobial properties and only few studies were conducted with fish samples. Combined effect of certain nanoparticles were also practiced in fish disease management (Zhang et al., 2021, 2022).

With this background, the present review is intended to focus on the role of different nanoparticles on the diagnosis of fish diseases and causative agents including the treatment of diseases. In addition, the effect of nanoparticles on the growth and development of pathogens in fish was discussed. Furthermore, this review highlights the role of nanoparticles in disease control which will be useful for fish culturists and aquaculture industries. Fig. 1 represents the schematic representation of the present study.

## 2. Fish diseases and management

It is a well-known fact that fish is the major contributor to ensuring global food and nutrition security (Ahmed et al., 2020). Diseases are considered as key universal problems faced by the fishes which cause mortality in turn reduce the yield. Pathogenic and non-pathogenic fish diseases are the two main categories that affect fish growth and development (El-Sayed Ali et al., 2014; Deenathayalan et al., 2023). Pathogenic fish diseases are considered to be highly perilous compared to non-pathogenic diseases. Non-pathogenic diseases are those that are not communicable to other fish and arise as a result of nutritional deficiencies, environmental pollutants, genetic anomalies, and widespread aeration, whereas pathogenic diseases are highly communicable to other fishes, resulting in mass mortality rates (Idowu et al., 2017; El-Sayed Ali et al., 2017). Pathogenic diseases occur due to four types of infectious agents, which include bacteria (*Aeromonas*, *Streptococcus*, *Pseudomonas*, *Vibrio*, etc., species), viruses (Rhabdovirus, Birnavirus, Iridovirus, Herpesvirus, Reovirus, Retrovirus, Orthomyxovirus, etc.), fungi (*Saprolegnia parasitica*, *Aphanomyces invadans*, *Fusarium oxysporum*, *Aspergillus*, *Batrachochytrium*, *Cladosporium*, etc., species) and parasites (*Gyrodactylus*, *Dactylogyrus*, *Ergasilus*, *Argulus*, *Caligus*, *Lernae*, etc., species) (Idowu et al., 2017; Abu-Elala et al., 2018b). Several fishes have specialized defense mechanisms against pathogens i.e., specific and non-specific defense (Chaplin, 2010). Specific defenses include adaptive immune responses against particular pathogens. Based on these defenses, vaccines are developed against certain pathogens (Somerset et al., 2005). However, non-specific defenses include mucus cells of the epidermis, skin, and scales which trap microbial organisms and arrest their growth (Sharma et al., 2012). The pathogens that succeed in breaking these barriers/defenses can be further killed by the inflammatory reactions produced by the fish through the transportation of increased blood flow and white blood cells to infected areas.

The fish immune system plays an important role in defending the fish against any disease-causing agent. Fishes have two types of immunity i.e. innate and adaptive. Innate immunity is the first line of defense against pathogen invasion, whereas adaptive immunity is much more pathogen-specific (Sharma et al., 2012). Non-specific cellular (eosinophils, cytotoxic cells, macrophages, and neutrophils) and humoral elements (C-reactive proteins, complement, interferons, lysozyme, transferrins, and lectins) make up innate im-

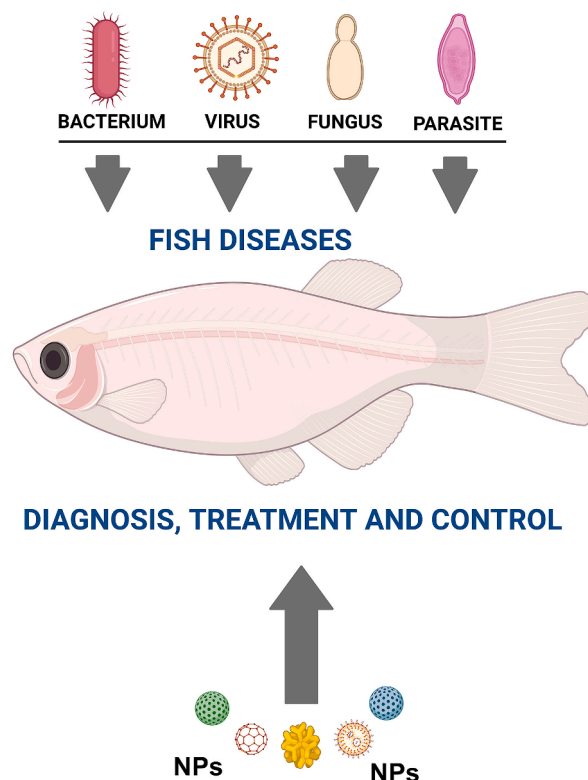


Fig. 1. Schematic representation of the present study.

munity, whereas adaptive immunity also includes humoral (IgM, IgD, and IgT antibodies) and cellular components (T-lymphocytes) that destroy pathogens (Firdaus-Nawi and Saad, 2016). To combat the invaded pathogens, both immune systems must function effectively together.

Fish health management has several requirements and needs to observe all aquaculture practices. Before that, fish farmers should be trained properly in aquaculture practices. Initially, it is a must to provide healthy seeds and young fish, proper nutrition, optimal water quality, hygienic conditions, and appropriate waste management practices (Anita and Khati, 2019). Fish diseases in the fisheries are managed by the usage of both synthetic and natural chemicals. Synthetic chemicals include antibiotics (sulfonamides and tetracyclines), vitamins, hormones, disinfectants (green malachite and hydrogen peroxide), and antihelminthics (pyrethroids and avermectin) (Citarasu, 2010). Natural chemicals include essential oils, herbal medicines, and probiotics. Generally, bacterial toxicity causes organ failure in localized or systemic infections (Kothari et al., 2019). There are various bacterial diseases in fish that can be controlled only through antibiotics. These antibiotics act as antimicrobial agents or chemo-medical agents (Romero et al., 2012). However, over usage of antibiotics may increase biological carrying in water and may also develop antibiotic resistance in bacteria. Probiotics are a natural method of treatment using beneficial microorganisms which survive in the digestive tract of fish to enhance growth, improve nutrient digestion, and inhibit pathogens. However, they raise histamine levels in fish and cause inflammation (Kothari et al., 2019).

Essential oils are plant-derived substances that have antibacterial, antifungal, and antiviral properties. These oils can be used as natural chemicals against a variety of pathogens (Mabrouk et al., 2021). Herbal medicines are similar types of products where different parts of plants (leaves, roots, barks, fruits, and seeds) are used as formulated extracts against fish pathogens (Citarasu, 2010; Anita and Khati, 2019). However, not all natural remedies may be regarded as advantageous due to drawbacks such as inadequate knowledge of their biological activity, mode of action, and management (Reverter et al., 2017). There is no sensible treatment for infections, thus all of these disease management techniques can only guarantee with an improvement in water quality. However, due to the effective use of vaccines against fish pathogens, aquaculture has emerged competently from disease outbreaks (Shah and Mraz, 2020). Vaccines have significantly reduced the use of antibiotics and the likelihood of drug resistance in aquaculture (Ma et al., 2019; Mondal and Thomas, 2022). In addition, maintaining the perfect records are essential in the aquaculture industry which covers all the aspects of farm operations. The recorded information helps to understand the nature of a disease outbreak and can easily diagnose the causative agents for their control. With the advancement of nanoscience, it is possible to manage the disease outbreaks in fish using safe nanoparticles.

### 3. Nanotechnology as a promising tool against fish diseases

Nanotechnology or nanoscience is proven to be one of the prospective fields appropriate to all fields of biology. Especially, the role of nanomaterials in biotechnology, biochemistry, microbiology, medicine, agriculture, and aquaculture is enormous. This technology is capable of examining, evaluating, manipulating, and fabricating objects on a nanometer scale (Can et al., 2011). Different nanoparticles are employed in aquaculture for various purposes such as enhancing fish development, water treatment, pond sanitation and removal of heavy metals, aquatic disease prevention, effective medicine, and nutrient delivery (Rajesh Kumar et al., 2008; Handy et al., 2012; Hua et al., 2012; Bhupinder, 2014). The use of nanotechnology in aquaculture may result in pollution and disease-free fish ponds, as well as assist in maintaining seafood safety measures during the packaging process (Rajesh Kumar et al., 2008; Bhattacharya et al., 2015; Bhat and Bhat, 2022). Mostly, gold, silver, copper oxide, zinc oxide, selenium, platinum, etc., nanoparticles are widely synthesized and used according to size, shape and delivery to a specific target. There are many nano-devices such as nanonets and nanosensors that are used to increase the pH and efficacy of water (Fajardo et al., 2022).

Controlling the disease outbreak is one of the biggest challenges to the aquaculture sector including fisheries. Nanodevices are the most prospective tools for detecting fish pathogens and monitoring fish diseases (Shaalan et al., 2016). The usage of nanostructure and nanosensor-based surfaces in aquaculture helps in the early detection of pathogens in water and delivers nanomedicine for effective therapy against pathogens (Muruganandam et al., 2019; Shah and Mraz, 2020). By integrating nanotechnology with biotechnology, nanovaccines have been designed which are more effective against pathogens than antibiotics. Due to the multifaceted benefits of nanoparticles to act against disease-causing pathogens in farmed fish, it is turned out to be a novel strategy (Bhattacharya et al., 2015). The use of nanoparticles in aquaculture resulted in the acceleration of the growth of shrimp or prawn post-larvae and the generation of pathogen-free seedlings (Muruganandam et al., 2019). The size, shape and chemical composition of nanoparticles, source and method of nanoparticle synthesis, concentration, exposure time, etc., are the key factors in fish disease management. Majorly, the nature (toxic or non-toxic) of nanoparticles is the valid concern before usage.

### 4. Various nanoparticles in fish disease diagnosis, treatment, and control

Nanoparticles or nanomaterials of gold, silver, copper, zinc, selenium, chitosan, etc., emerged as potential alternatives to antibiotics (Swain et al., 2014; Pati et al., 2021). Moreover, they qualified as next-generation nanomedicine because of their unique medicinal properties and cutting-edge pharmacological capabilities (De Villiers et al., 2008; Bharti and Singh, 2009; Das et al., 2013). Table 1 represents the nanoparticles used for different fish disease diagnosis, treatment, and control.

Gold nanoparticles are frequently used because of their size, inherent colloidal stability, and great number of coatings that can easily be attached. Previously, gold NPs were considered to be bioinert compounds, characterized by low toxicity (Velmurugan et al., 2014; Pierre et al., 2024). Due to their significant absorption and scattering properties, gold nanoparticles have distinctive optical features. Presently, they are rigorously used in different domains including cancer research (Tu et al., 2020). Saleh et al. (2011) detected *Aeromonas salmonicida* as a causative agent of furunculosis in both salmonid and non-salmonid fishes through antibody-coated gold

**Table-1**

Various nanoparticles involved in fish disease management.

| Pathogen        | Name                                                                                                                                                                                                                                                                                          | Fish                                              | NPs             | Mode of action                                                                               | Reference                    |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|-----------------|----------------------------------------------------------------------------------------------|------------------------------|
| <b>Bacteria</b> | <i>Streptococcus iniae</i> , <i>Lactococcus garvieae</i> , <i>Yersinia ruckeri</i> and <i>Aeromonas hydrophila</i>                                                                                                                                                                            | Fish                                              | Silver          | Showed the antibacterial activity for certain species.                                       | Soltani et al. (2009)        |
|                 | <i>Aeromonas salmonicida</i>                                                                                                                                                                                                                                                                  | Salmonid and non-salmonid fishes                  | Gold            | Identification of pathogen                                                                   | Saleh et al. (2011)          |
|                 | <i>Lactococcus garvieae</i> and <i>Streptococcus siniae</i>                                                                                                                                                                                                                                   | Rainbow trout                                     | Silver          | Exhibited antibacterial activity                                                             | Mehdi and Ansari (2011)      |
|                 | <i>Aeromonas hydrophila</i>                                                                                                                                                                                                                                                                   | Fishes                                            | Silver          | Demonstrated antibacterial activity                                                          | Mahanty et al. (2013)        |
|                 | <i>Bacterial species</i>                                                                                                                                                                                                                                                                      | Fish                                              | CuO             | Showed the antibacterial activity                                                            | Swain et al. (2014)          |
|                 | <i>Aeromonas hydrophila</i> -MTCC 646, <i>Aeromonas bestiarum</i> -MTCC 10106, <i>Pseudomonas fluorescens</i> -MTCC 67 and <i>Edwardsiella tarda</i> -MTCC 2400                                                                                                                               | Fishes                                            | Gold            | Isolated from Cashew nut shell liquid and showed antibacterial activity for certain species. | Velmurugan et al. (2014)     |
|                 | <i>Aeromonas hydrophila</i> , <i>Pseudomonas fluorescens</i> , and <i>Flavobacterium branchiophilum</i>                                                                                                                                                                                       | Fish                                              | CuO             | Nanoparticles acted as antibacterial agent                                                   | Kumar et al. (2015)          |
|                 | <i>Aeromonas hydrophila</i> , <i>Edwardsiella ictaluri</i> , <i>Aeromonas salmonicida</i> , <i>Edwardsiella tarda</i> , <i>Francisella noatunensis</i> and <i>Yersinia ruckeri</i>                                                                                                            | Catfish ( <i>Ictalurus punctatus</i> ) and fish   | Silver          | Exhibited the antibacterial function for certain species                                     | Shaaan et al. (2017)         |
|                 | <i>Aeromonas hydrophila</i> , <i>Edwardsiella ictaluri</i> , <i>Aeromonas salmonicida</i> , <i>Edwardsiella tarda</i> , <i>Francisella noatunensis</i> and <i>Yersinia ruckeri</i>                                                                                                            | Fish and Catfish ( <i>Ictalurus punctatus</i> )   | ZnO             | Showed the antibacterial activity for some species.                                          | Shaaan et al. (2017)         |
|                 | <i>Aeromonas salmonicida</i>                                                                                                                                                                                                                                                                  | <i>Oncorhynchus mykiss</i> (Rainbow trout)        | Silver          | Showed the antibacterial function.                                                           | Shaaan et al. (2018)         |
|                 | <i>Aeromonas hydrophila</i>                                                                                                                                                                                                                                                                   | <i>Oreochromis mossambicus</i>                    | ZnO             | Exhibited antibacterial activity                                                             | Anjugam et al. (2018)        |
|                 | <i>Aeromonas hydrophila</i>                                                                                                                                                                                                                                                                   | Fish                                              | Copper          | Acted as antibacterial agent                                                                 | Abu-Elala et al. (2018a)     |
|                 | <i>Aeromonas hydrophila</i> , <i>Aeromonas sobria</i> , <i>Streptococcus agalactiae</i>                                                                                                                                                                                                       | <i>Oreochromis niloticus</i>                      | Chitosan        | Antagonistic action against certain bacteria                                                 | Abdel-Razek (2019)           |
|                 | <i>Staphylococcus aureus</i> and <i>Pseudomonas fluorescens</i>                                                                                                                                                                                                                               | <i>Oreochromis niloticus</i>                      | ZnO             | Showed the antibacterial activity.                                                           | Awad et al. (2019)           |
|                 | <i>Aeromonas</i> , <i>Pseudomonas</i> , <i>Edwardsiella</i> , <i>Yersinia</i> , <i>Francisella</i> , <i>Citrobacter</i> and <i>Vibrio</i>                                                                                                                                                     | Fish species                                      | Chitosan        | Exhibited the antibacterial activity.                                                        | Ahmed et al. (2019)          |
|                 | <i>Escherichia coli</i> , <i>Streptococcus mutants</i> , <i>Staphylococcus aureus</i> , <i>Serratia marcescens</i> , <i>Candida albicans</i> , <i>Micrococcus luteus</i> , <i>Salmonella typhi</i> , <i>Pseudomonas fluorescens</i> , <i>Proteus</i> species and <i>Mycobacterial</i> species | Marine fish and <i>Mycobacterium tuberculosis</i> | Gold            | Exhibited antimicrobial and anti mycobacterial activities for certain species.               | Rajasekar et al. (2020)      |
|                 | <i>Pseudomonas aeruginosa</i>                                                                                                                                                                                                                                                                 | African catfish <i>Clarias gariepinus</i> .       | ZnO             | Worked as antibacterial agent                                                                | Mahboub et al., 2020         |
|                 | <i>Vibrio harveyi</i> , <i>Vibrio parahaemolyticus</i> , <i>Vibrio anguillarum</i> and <i>Vibrio vulnificus</i>                                                                                                                                                                               | Fish                                              | Silver          | Showed the antibacterial activity.                                                           | Fatima et al. (2020)         |
|                 | <i>Aeromonas veronii</i>                                                                                                                                                                                                                                                                      | <i>Xiphophorus hellerii</i>                       | ZnO             | Showed better antibacterial activity than silver NPs                                         | Das et al. (2020)            |
|                 | <i>Aeromonas hydrophila</i>                                                                                                                                                                                                                                                                   | <i>Oreochromis niloticus</i>                      | Silver          | Exhibited antibacterial activity                                                             | Mabrouk et al. (2021)        |
|                 | <i>Aeromonas veronii</i> strain and <i>Stenotrophomonas maltophilia</i> strains                                                                                                                                                                                                               | Freshwater fish                                   | ZnO             | Acted as antibacterial agent                                                                 | Pati et al. (2021)           |
|                 | <i>Aeromonas hydrophila</i> , <i>Pseudomonas aeruginosa</i> , <i>E. coli</i> , <i>Aeromonas</i> , <i>Enterococcus</i> , <i>Salmonella</i> , and <i>Shigella</i>                                                                                                                               | Fish species                                      | Selenium NPs    | Worked as antibacterial agents for certain species.                                          | Dash et al. (2022)           |
|                 | <i>Streptococcus agalactiae</i> , <i>Aeromonas hydrophila</i> and <i>Vibrio alginolyticus</i>                                                                                                                                                                                                 | Fish                                              | Silver          | Act as antibacterial agent                                                                   | Ghetas et al. (2022)         |
|                 | <i>Aeromonas hydrophila</i>                                                                                                                                                                                                                                                                   | <i>Oreochromis mossambicus</i>                    | Se NPs with ZnO | Exhibited the antibacterial activity                                                         | Yazhiniprabha et al. (2022). |
|                 | <i>Yersinia ruckeri</i>                                                                                                                                                                                                                                                                       | Rainbow trout                                     | Copper          | Worked as an antibacterial agent                                                             | Delavari et al. (2022)       |
|                 | <i>Aeromonas salmonicida</i>                                                                                                                                                                                                                                                                  | Zebrafish                                         | Carbon-Ag NPs   | Acted as an antibacterial agent                                                              | Zhang et al. (2022)          |
|                 | <i>Aeromonas hydrophila</i>                                                                                                                                                                                                                                                                   | <i>Labeo rohita</i>                               | Silver          | Demonstrated the antibacterial activity                                                      | Popoola et al. (2023)        |

(continued on next page)



Table-1 (continued)

| Pathogen | Name                                                                                                                                            | Fish                                                  | NPs                         | Mode of action                                                     | Reference                     |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-----------------------------|--------------------------------------------------------------------|-------------------------------|
| Parasite | <i>Aeromonas hydrophila</i> and <i>Vibrio parahaemolyticus</i>                                                                                  | <i>Oreochromis mossambicus</i> .                      | ZnO                         | Acted as an antibacterial agent                                    | Abinaya et al. (2023)         |
|          | <i>Aeromonas hydrophila</i>                                                                                                                     | Nile Tilapia                                          | Platinum along with Lec     | Functioned as antibacterial agent                                  | Lakshmi et al. (2023)         |
|          | <i>Aeromonas hydrophila</i>                                                                                                                     | <i>Ctenopharyngodon idella</i>                        | Chitosan                    | Worked as an antibacterial agent                                   | Zhang et al. (2023)           |
|          | <i>Aeromonas hydrophila</i>                                                                                                                     | <i>Oreochromis niloticus</i>                          | ZnO                         | Acted as antibacterial agent                                       | Sherif et al. (2023)          |
|          | <i>Streptococcus agalactiae</i>                                                                                                                 | <i>Oreochromis niloticus</i>                          | Se NPs                      | Acted as an antibacterial agent                                    | Ni et al. (2023)              |
|          | <i>Heterosporis saurida</i>                                                                                                                     | <i>Saurida undosquamis</i>                            | Gold                        | Act as antimicrosporidial agent in lizardfish                      | Saleh et al. (2016)           |
|          | <i>Ichthyophthirius multifiliis</i> - Protozoan                                                                                                 | <i>Carassius auratus</i>                              | Silver                      | Exhibited the antiprotozoan effect                                 | Daniel (2016)                 |
|          | <i>Ichthyophthirius multifiliis</i>                                                                                                             | <i>Oncorhynchus mykiss</i>                            | Gold                        | Act as an antiprotozoal agent in rainbow trout                     | Saleh et al. (2017)           |
|          | <i>Lernaea cyprinacea</i>                                                                                                                       | Goldfish                                              | Chitosan-Silver composition | Exhibited antiparasitic effect                                     | Abu-Elala et al. (2018b)      |
|          | <i>Cichlidogyrus</i> species                                                                                                                    | Fish                                                  | Silver                      | Showed antibacterial activity                                      | Pimentel-Acosta et al. (2019) |
| Virus    | <i>Gyrodactylus kobayashi</i> .                                                                                                                 | Goldfish                                              | Gold                        | Fight against monogenean parasite <i>Gyrodactylus kobayashi</i>    | Tu et al. (2020)              |
|          | Nervous necrosis virus (NNV)                                                                                                                    | Fish stocks                                           | Gold                        | Detection of virus                                                 | Hassantabar et al. (2021)     |
|          | Noda virus                                                                                                                                      | <i>Dicentrarchus</i> (D.) <i>labrax</i>               | Gold                        | Detection of nervous necrosis virus particles                      | Toubanaki et al. (2020)       |
|          | <i>Rana</i> virus                                                                                                                               | Largemouth bass fish ( <i>Micropterus salmoides</i> ) | Gold                        | Detection of largemouth bass virus                                 | Jin et al. (2020)             |
|          | Noda virus                                                                                                                                      | <i>Sparus aurata</i> and <i>Dicentrarchus labrax</i>  | Gold                        | Nanoparticles were used for identification of conserved RNA region | Cherif et al., 2021           |
| Fungus   | <i>Aphanomyces invadans</i>                                                                                                                     | Fish                                                  | Gold                        | Designed for gold nano particles-based-DNA sensor                  | Kuan et al. (2013)            |
|          | <i>Aphanomyces invadans</i>                                                                                                                     | <i>Carassius auratus</i>                              | Silver                      | Showed the antifungal activity                                     | Daniel (2016)                 |
|          | <i>Aphanomyces invadans</i>                                                                                                                     | Fish                                                  | Silver                      | Exhibited the antifungal activity.                                 | Shaaan et al. (2017)          |
|          | <i>Aphanomyces invadans</i>                                                                                                                     | Fish                                                  | ZnO                         | Worked as antifungal agent                                         | Shaaan et al. (2017)          |
|          | <i>Aspergillus flavus</i> , <i>Mucor</i> , <i>Candida</i> species, <i>Aspergillus niger</i> , <i>A. fumigatus</i> , and <i>Fusarium</i> species | <i>Oreochromis niloticus</i>                          | Chitosan                    | Acted as antifungal agent for certain fungi                        | Abdel-Razek, 2019             |
|          | <i>Saprolegnia parasitica</i>                                                                                                                   | Fish                                                  | Copper related CCCSN        | Act as antifungal agent                                            | Zhang et al. (2021)           |
|          | <i>Aspergillus flavus</i> , <i>Fusarium moniliforme</i> and <i>Candida albicans</i>                                                             | Fish                                                  | Silver                      | Exhibited antifungal activity                                      | Ghetas et al. (2022)          |

nanoparticle immunoassay. Kuan et al. (2013) designed a gold nanoparticles-based DNA sensor for the detection of *Aphanomyces invadans*, a fungal pathogen of fish. Both gold and silver nanoparticles were synthesized from cashew nut shell liquid which showed antibacterial activities (Velmurugan et al., 2014). Bacterial strains such as *Aeromonas hydrophila* MTCC 646, *Aeromonas bestiarum* MTCC 10106, *Pseudomonas fluorescens* MTCC 67, and *Edwardsiella tarda* MTCC 2400 were used for antibacterial assay. It has been established that AuNP-based assay is a crucial laboratory diagnostic tool for the quick identification of unamplified nucleic acids in hybridization tests (Saleh et al., 2015).

Metal nanoparticles can be treated as a substitute for new antimicrosporidial drugs production to combat diseases in fish industries. Gold nanoparticles effectively used as a potential antimicrosporidial agent against microsporidia, *Heterosporis saurida* to reduce disease in lizardfish (*Saurida undosquamis*) (Saleh et al., 2016). Gold NPs were used as a potential antiprotozoal agent against *Ichthyophthirius multifiliis* to reduce protozoan disease in rainbow trout (*Oncorhynchus mykiss*) (Saleh et al., 2017). Rajasekar et al. (2020) synthesized gold nanoparticles using extracellular metabolites of fish gut microorganisms and used for antibacterial and antimycobacterial activities. Further, to test antimicrobial activity, *Escherichia coli*, *Streptococcus mutants*, *Staphylococcus aureus*, *Serratia marcescens*, *Candida albicans*, *Micrococcus luteus*, *Salmonella typhi*, *Pseudomonas fluorescens*, and *Proteus* species were used. Tu et al. (2020) employed gold nanoparticles as new solid support to identify drug targets of the natural bioactive compound arctigenin (ARG) against fish monogenean parasite *Gyrodactylus kobayashi*. Detection of largemouth bass virus (LMBV) was carried out using probe-modified gold nanoparticle as a colorimetric biosensor (Jin et al., 2020). Toubanaki et al. (2020) detected the fish noda virus (nervous necrosis virus) particles through a gold nanoparticle-based lateral flow strip biosensor. Chérif et al. (2021) designed a method enabling the direct detection of a conserved RNA region within the capsid protein gene of nodavirus of fishes. Moreover, they validated this work using extracted RNA samples isolated from healthy *Sparus aurata* and clinically infected *Dicentrarchus labrax* species. Anti-

body-gold nanoparticles specific for nervous necrosis virus (NNV) were used for the detection of the virus in fish stocks through lateral flow immunochromatography strip test (Hassantabar et al., 2021) (Table 1).

In contrast, toxic morphological malformation of zebrafish (*Danio rerio*) embryos was noticed when exposed to functionalized gold NPs with a high concentration (625  $\mu\text{g mL}^{-1}$ ) of triphenylphosphine monosulfonate (TPPMS), whereas no toxic effect was noticed in functionalized gold NPs with low concentration (62.5  $\mu\text{g mL}^{-1}$ ) of TPPMS (Bar-Ilan et al., 2009). Dedeh et al. (2015) proved the toxic effects of Au NPs when compared to ionic gold particles in zebrafish and found alteration in expression patterns of genes involved in DNA repair, oxidative stress, mitochondrial metabolism, etc. Both *in vitro* and *in vivo* studies proved that gold nanoparticles are involved in antioxidant system of fish (Sani et al., 2021). d'Amora et al. (2021) proved that gold NPs altered the growth of embryo and larvae and caused histological damage of both liver and ovary in adult zebrafish. Recently, Windell et al. (2023) proved that multi-size or modified gold nanoparticles are less toxic in zebrafish. This data indicates that the toxicity levels alter through modification of nanoparticles.

Silver nanoparticles are the major group which is having many applications (Shaalán et al., 2017). Specifically, silver NPs are used in agricultural, aquaculture, and biomedical applications due to their unique biological, optical, and thermal properties, which allow them to be used in a variety of biological and chemical sensors (Márquez et al., 2018; Khursheed et al., 2023). They contain high electrical and thermal conductivity, chemical stability, catalytic activity, scattering, and nonlinear optical performance. Bacterial disease is a major concern in aquaculture and the use of antibiotics can reduce the effect. However, these antibacterial drugs pose a serious threat to the aquatic environment which in turn provokes resistance against bacterial strains (Tuševljak et al., 2013). This effect can be curbed with the application of potential nanoparticles such as silver NPs. Silver nanoparticles are one of the most effective agents because they have good antimicrobial efficacy against some bacteria, viruses and other eukaryotic microorganisms like protozoans (Pelgrift and Friedman, 2013). Soltani et al. (2009) used *Streptococcus iniae*, *Lactococcus garvieae*, *Yersinia ruckeri* and *Aeromonas hydrophila* for antibacterial activity. Similarly, silver nanoparticles exhibited antibacterial activity against *Lactococcus garvieae* and *Streptococcus siniae* obtained from diseased rainbow trout (Mehdi and Ansari, 2011). Silver nanoparticles isolated from mangrove plant exhibited antibacterial activity (Umashankari et al., 2012). They worked against *Pseudomonas fluorescens*, *Proteus*, and *Flavobacterium* species isolated from infected marine ornamental fish, *Dascyllus trimaculatus* (Table 1). Mahanty et al. (2013) extracted silver nanoparticles from different plant sources such as mango, papaya, banana, and eucalyptus which inhibit *Aeromonas hydrophila*, a bacterial fish pathogen. Specifically, silver NPs synthesized from papaya leaf extract had maximum antimicrobial activity. Daniel (2016) worked on the effects of silver nanoparticles on the red spot (caused by *Aphanomyces invadans*) and white spot (caused by *Ichthyophthirius multifiliis*) diseases of *Carassius auratus* (ornamental goldfish). They found the protective effects of silver nanoparticles against these protozoan and fungal diseases of fish.

Shaalán et al. (2017) used silver nanoparticles against fish pathogen bacteria such as *Aeromonas hydrophila*, *Edwardsiella ictaluri*, *Aeromonas salmonicida*, *Edwardsiella tarda*, *Francisella noatunensis*, and *Yersinia ruckeri* to test the antibacterial assay and succeeded. Also, the antifungal activity against *Aphanomyces invadans* with the same silver nanoparticles was noticed. The antiparasitic effect of silver NPs on the protozoan *Ichthyophthirius multifiliis*, which causes white spot syndrome, was investigated by Saleh et al. (2017). Shaalan et al. (2018) also emphasized the role of AgNPs as antibacterial agents against *Aeromonas salmonicida* subspecies *Salmonicida* which causes the disease in rainbow trout. The effect of silver nanoparticles against monogenean parasites of fish was noticed (Pimentel-Acosta et al., 2019). They used both adults and eggs of *Cichlidogyrus* parasitic species for the experiments. Silver nanoparticles were synthesized from red alga (*Portieria hornemannii*) showed antibacterial activity against *Vibrio harveyi*, *Vibrio parahaemolyticus*, *Vibrio anguillarum*, and *Vibrio alginolyticus* (Fatima et al., 2020). Mabrouk et al. (2021) noticed the antibacterial efficacy using silver nanoparticles using *Oreochromis niloticus* challenged with *Aeromonas hydrophila* (Table 1). In addition, they observed the toxicity of Ag NPs at higher doses. Silver nanoparticles were synthesized using both chemical and biological (leaves of *Origanum vulgare*) methods and tested against fish pathogens (Ghetas et al., 2022). Further, these nanoparticles exhibited both antibacterial (against *Streptococcus agalactiae*, *Aeromonas hydrophila*, and *Vibrio alginolyticus*) and antifungal (against *Aspergillus flavus*, *Fusarium moniliforme*, and *Candida albicans*) activities. Silver nanoparticles were used in rohu (*Labeo rohita*) cultures to fight against *Aeromonas hydrophila* and succeeded (Popoola et al., 2023).

On the otherhand, the toxicity of silver NPs was already studied and the action of Ag NPs on diverse aquatic organisms has been reported by several researchers (Popoola et al., 2023). Specifically, silver NPs induces the developmental toxicity, genotoxicity and immunotoxicity. Gagné et al. (2012) noticed different gene alterations in *Oncorhynchus mykiss* when exposed to silver nanoparticles and dissolved silver. Khan et al. (2015) emphasized the silver toxicity in fish and consequences. Silver nanoparticles exposure activated the enzymatic antioxidant system in the liver of *Prochilodus lineatus* (Ale et al., 2018). Silver nanoparticles showed toxicity in zebrafish and influenced their hatching rate, heartbeat rate and swimming activities (d'Amora et al., 2021). Prolonged exposure of fish to high concentration of silver NPs induced histological alteration and fatality in *Cyprinus carpio* (Kakakhel et al., 2021). Marinho et al. (2021) observed the toxic effects of silver nanoparticles in zebrafish brain, muscle, gill and liver tissues and modifications in their antioxidant system. Younas et al. (2022) proved that green synthesized silver NPs showed less toxicity in fish when compared to chemically synthesized silver NPs. This data indicates that both the source and method of synthesis of nanoparticles are also key factors to avoid toxicity.

Generally, copper performs a range of functions, such as acting as a cofactor for several enzymes, and involved in cell signaling pathways, defense against free radicals, and activity of immune cells (Ren et al., 2009). However, elevated copper concentrations are highly toxic and the toxicity cannot be related to one specific mode of action. Copper is very minutely present in both humans and plants, where it helps to control a variety of biological and metabolic processes (Al-Hakkani, 2020). Among the metal oxide nanoparticles, copper oxide nanoparticles have drawn a lot of interest due to their diverse properties and potential uses in a variety of fields, most notably nanomedicine and the biomedical sciences (Mohamed, 2020). Copper nanoparticles are renowned for being inexpen-

sive, physically and chemically stable, and simple to combine with polymers (Chandraker et al., 2020). Copper nanoparticles have antibacterial, antifungal, antiviral, and anticancer properties. Both gram-positive and gram-negative bacteria, including *Escherichia coli*, *Staphylococcus aureus*, and *Bacillus subtilis*, were susceptible to copper nanoparticles as per the studies of Ruparelia et al. (2008).

Swain et al. (2014) used copper oxide nanoparticles as antibacterial agents to control certain fish diseases. Kumar et al. (2015) synthesized CuO nanoparticles from *Aloe vera* extract and used as antibacterial agents to fight against certain fish bacterial pathogens. They used *Aeromonas hydrophila*, *Pseudomonas fluorescens*, and *Flavobacterium branchiophilum* for antibacterial assay. Abu-Elala et al. (2018a) used copper nanotubes as antimicrobial agents to cure fish diseases. Particularly, they used *Aeromonas hydrophila* for antibacterial activity which causes disease in fish. Zhang et al. (2021) controlled the saprolegniosis disease in fish caused by a fungal species namely *Saprolegnia parasitica* through copper/carbon core/shell nanoparticles (CCCSNs). Delavari et al. (2022) used copper nanoparticles for disease resistance studies against *Yersinia ruckeri* in rainbow trout (*Oncorhynchus mykiss*) and succeeded (Table 1).

In contrast, both copper and copper oxide NPs were toxic to the aquatic organisms particularly algae and crustaceans at higher doses. At elevated concentration, copper NPs may induce formation of ROS and activate antioxidant enzymes in turn alter the cell growth. Vajargah et al. (2018) reported the toxic effects of copper oxide NPs in *Poecilia reticulata* (Guppy fish) and observed increased mortality and less reproductive success. Malhotra et al. (2020) emphasized that copper nanoparticles are less toxic than normal copper sulfate when treated with fish tissues. This work indicates the importance of nanoparticles in fish disease management. Copper oxide NPs cause oxidative stress, genotoxicity, immunotoxicity, cytotoxicity, neurotoxicity and inflammation in different cell lines including fish at high concentration (Sajjad et al., 2023; Balasubramanian et al., 2024).

Zinc is an essential nutrient for aquatic organisms which is a sensitive element. In addition, zinc oxide nanoparticles have surfaced as the highly potential metal-based nano drugs due to their biocompatibility, higher specific surface area, higher bioavailability, molecular dispersion, and antimicrobial properties (Swain et al., 2016; Raje et al., 2018). Zinc oxide nanoparticles have been successfully synthesized using various plant extracts (Singh et al., 2018). Zinc oxide NPs have the potential to perform multifunctional activities in aquaculture including disease control. Specifically, zinc oxide nanoparticles were used against fish pathogen bacteria such as *Aeromonas hydrophila*, *Edwardsiella ictaluri*, *Aeromonas salmonicida*, *Francisella noatunensis*, *Edwardsiella tarda*, and *Yersinia ruckeri* to test the antibacterial activity and succeeded (Shaalan et al., 2017). In addition, antifungal activity against *Aphanomyces invadans* with the same nanoparticles was observed. Zinc oxide NPs are widely used as feed additives to boost growth and immunity in *Ctenopharyngodon idella* and African catfish fingerlings (Faiz et al., 2015; Onuegbu et al., 2018).

There have been few investigations into the antibacterial and antifungal effects of green ZnO NPs on aquatic diseases (Chandran et al., 2018). Effect of  $\beta$ -1,3 glucan binding protein-based ZnO nanoparticles improved disease resistance in *Oreochromis mossambicus* against *Aeromonas hydrophila* (Anjugam et al., 2018). Awad et al. (2019) proved that ZnO nanoparticles improved the disease resistance in *Oreochromis niloticus* against *Aeromonas sobria*. Zinc oxide nanoparticles exhibited better results when compared to silver nanoparticles in the management of *Aeromonas veronii* infection in *Xiphophorus hellerii* (Das et al., 2020). Mahboub et al. (2020) proved the efficacy of nano zinc oxide in disease resistance against *Pseudomonas aeruginosa* using the African catfish *Clarias gariepinus* (Table 1). Pati et al. (2021) observed the disease control in freshwater fish caused by *Aeromonas veronii* strain ONKP1 (MN602971) and *Stenotrophomonas maltophilia* strain ONKP2 (MN602972) through ZnO nanoparticles. They prepared the nanoparticles from *Aloe vera* plant in green synthesis process. Abinaya et al. (2023) found that ZnO nanoparticles are involved in disease resistance against *Aeromonas hydrophila* and *Vibrio parahaemolyticus* in *Oreochromis mossambicus*. Zinc oxide nanoparticles improved the disease resistance in *Oreochromis niloticus* against *Aeromonas hydrophila* (Sherif et al., 2023).

In contrast, the toxicity of zinc oxide nanoparticles on various microorganisms and fish has been demonstrated by several researchers (Hao et al., 2013). Rajkumar et al. (2022) observed sub-acute toxicity in *Cyprinus carpio* and found oxidative stress when exposed to zinc oxide NPs. Deenathayalan et al. (2023) proved the lethal effects of ZnO NPs in zebrafish embryos and observed the oxidative stress. The disruption of the cell membrane upon interaction with ZnO NPs has also been attributed to the cause of growth inhibition of cells. Zinc oxide NPs can be part of nutrition at low concentrations, but becomes toxic when over accumulated (Mandal et al., 2024).

Selenium is an essential trace element that is required for normal metabolic activities and body functions (Azdi et al., 2013). Selenium enhances the immune system, physiology, growth, and fertility of the animals and is also toxic in larger amounts. Dash et al. (2022) isolated selenium nanoparticles from *Blumea axillaris* plant species and found antibacterial activity. They used *Aeromonas hydrophila*, *Pseudomonas aeruginosa*, *E. coli*, *Aeromonas*, *Enterococcus*, *Salmonella*, and *Shigella* species for antibacterial assay. Selenium nanoparticles along with zinc oxide fight against the aquatic pathogen *Aeromonas hydrophila* in *Oreochromis mossambicus* (Yazhini Prabha et al., 2022). Selenium NPs were prepared using polysaccharide-protein complexes which exhibited antibacterial activity against *Streptococcus agalactiae* in *Oreochromis niloticus* (Ni et al., 2023).

On the other hand, selenium NPs are toxic in high doses and the generation of nanoparticles from various sources has been a valid concern. Saffari et al. (2016) showed that Se NPs act more efficiently on growth performance and antioxidant defense system of common carp than organic and inorganic sources of selenium. This data indicates the importance of nanomaterials in fish disease management. Additionally, Juhász et al. (2017) showed that the moderate level of selenium enrichment influences the rearing efficiency of fish larvae, however higher concentrations cause adverse effects. Kumar et al. (2018) found that both selenium and selenium NPs exhibited deleterious effects on *Pangasius hypophthalmus* fish growth and alters the antioxidant defense systems. Fan et al. (2022) observed different toxic levels in the gills and liver of zebrafish when exposed to selenium NPs which indicates the organ specificity. Hariharan et al. (2024) noticed that selenium NPs synthesized through biological methods showed less toxicity in zebrafish embryos.

Chitosan is a naturally occurring, biodegradable polysaccharide extracted from chitin (Ahmed et al., 2020). Chitosan NPs are currently employed as an efficient antibacterial and antifungal agent in *Oreochromis niloticus* (Abdel-Razek, 2019). They also observed antibacterial and antifungal properties for certain species. Ahmed et al. (2019) worked with chitosan nanoparticles and proved its an-



tibacterial activities against *Aeromonas*, *Pseudomonas*, *Edwardsiella*, *Yersinia*, *Francisella*, *Citrobacter*, and *Vibrio* species which are causing severe diseases in fish (Table 1). Zhang et al. (2023) applied carboxymethyl chitosan nanoparticles as antibacterial agents against *Aeromonas hydrophila* in *Ctenopharyngodon idella*. Abu-Elala et al. (2018b) used chitosan-silver nanocomposites to control *Lernaecyprinae* (copepod parasite) infestation in goldfish aquaria. On the other hand, toxicity evaluation of chitosan nanoparticles was carried out by Hu et al. (2011) using zebrafish embryo model. Zoe et al. (2023) emphasized that chitosan toxicity in zebrafish depends on the concentration, duration of exposure, pH of the solution etc.

Although in recent years, many studies were conducted using carbon nanoparticles, less studies were found on management of fish diseases. Carbon-silver nanocomposites exhibited antibacterial activity and enhanced the resistance against *Aeromonas salmonicida* in zebrafish (Zhang et al., 2022). Platinum is a rare and valuable metal that has been widely employed in various fields including biochemistry, biotechnology, biomedical sciences etc. Very recently, Lakshmi et al. (2023) demonstrated the rescuing *Aeromonas hydrophila*-infected Nile Tilapia using Lec-Pt nanoparticles. In contrast, Demir et al. (2020) evaluated the cytotoxic nature of platinum ions and platinum NPs and proved that the PVP (polyvinylpyrrolidone) coated Pt NPs exhibited no toxic nature. This data indicates that addition of metals or compounds to nanoparticle reduces the toxicity.

Moreover, treatment of water with nanoparticles in the aquaculture industry showed the positive results by suppressing the growth of pathogens (Hamad, 2019). Further, use of nanoparticles for drug delivery in fish against pathogens was established (Fenaroli et al., 2014). In contrast, negative effects of combination of nanoparticles on fishes were also reported by Abu-Elala et al. (2018b). By observing various useful effects reported above, nanotechnology is a blessing for sustainable aquaculture because it can overcome major obstacles including increased disease susceptibility, the advent of novel pathogens, and the poor efficacy of current diagnostic methods in the treatment of diseases.

## 5. Future prospects

Fish industry is an important sector of food production, providing nutritional security to the food basket, contributing to agricultural exports, and engaging millions of people in different activities. With diverse aquatic resources, any country can exhibit a sustained increment in fish production. In addition, the rapid growth of the human population has increased the demand for fish and fish products. However, the fisheries sector is facing severe setbacks in the form of disease outbreaks. But the management of fisheries is a highly difficult process. By keeping the importance, we highlighted the role of different nanoparticles in fish disease management. Specially, the application of nanoparticles for diagnosis, treatment, and control of fish pathogens increased in recent times. Selection of the element or chemical for nanoparticle preparation is one of the key points. Production of nanoparticles from various sources is an important concern for disease management practice. Generally, synthesis from green and other eco-friendly methods will give better results. Precautions at the culture level are a valid point in the aquaculture sector. Water treatment with suitable and safe nanoparticles in aquaculture industries will be highly helpful for the maintenance of cultures.

Advanced technology for disease control is one of the major areas in the aquaculture sector. Particularly, using of nanoparticles for the diagnosis and treatment of fish diseases must be given priority. Novel and conjugated nanoparticles may be useful for rapid and specific detection of fish pathogens. The preparation of useful and non-toxic nanoparticles is one of the concerns in controlling fish diseases. Biomedical suggestions such as drug delivery in diseased fish using appropriate nanoparticles will be useful to avoid various diseases. In addition, using of nanotubes, nanorods, nanowires, nanocapsules, etc., will give better response in detecting and controlling the fish diseases. The method of nanoparticle administration is one of the key points to consider. Using of nanoparticles in vaccine development is also a considerable point in fish medicine. Choice of the nanoparticle is one of the main concerns at present scenario due to side effects or toxic nature. In general, toxic nature of nanoparticles depends on size, shape, concentration, surface charge, source and mode of synthesis, route of administration, time of exposure, immune levels of target species etc. Specifically, both concentration or dose of nanoparticle and exposure time are the key factors to avoid toxic effects. Fig. 2 illustrates nanoparticle usage cautions for the detection and controlling the fish diseases. Maintenance of perfect records is essential in the fish industry which will give accurate information on disease outbreaks and their control methods. Training on awareness of nanotechnology for the fish culturists will improve the aquaculture sector. Overall, maintaining the hygiene conditions in the aquaculture industry is the ultimate solution for sustainable fish yield.

## 6. Conclusions

Fish is a vital source of protein, PUFAs, minerals, and vitamins which gained a lot of commercial value around the world. However, their susceptibility to various diseases caused by bacteria, viruses, fungi, and parasites made them a source of concern in intensified aquaculture due to economic loss. Over usage of antibiotics by aquafarmers against these pathogens pollutes the fish and its environment. Hence, this review has emphasized the role of various nanoparticles in the diagnosis of fish diseases, treatment, and control methods. Specifically, it highlights the fact that nanotechnology is emerging as a promising means of controlling various diseases in fish. In contrast, this review highlighted the toxicity of nanoparticles in fish disease management. Overall, the information of this review will be useful for fish culturists and aquaculture industries.

## Funding

There is no specific funding for this review article.

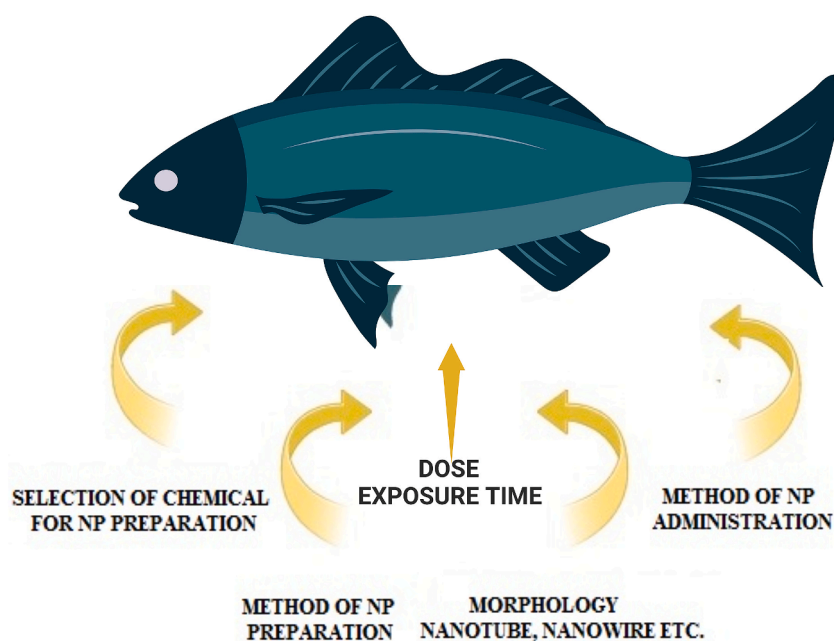


Fig. 2. Fish disease control by nanoparticles.

#### CRediT authorship contribution statement

**Rajeswari Dasari:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Anu Prasanna Vankara:** Writing – review & editing, Writing – original draft, Supervision, Formal analysis, Data curation, Conceptualization. **Riazunnisa Khateef:** Writing – review & editing, Writing – original draft, Software. **Venkata Ravi Teja Shegu:** Writing – review & editing, Software. **Vijayalakshmi Degati:** Writing – review & editing. **Chandrasekhar Thummala:** Writing – review & editing.

#### Declaration of competing interest

Here we are submitting the manuscript entitled ‘Role of nanoparticles in fish disease management: A review’ to your esteemed Biocatalysis and Agricultural Biotechnology as review article.

All the authors declared that they have no conflict of interest.

#### Data availability

The data that has been used is confidential.

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